

Decoding the Genomic Language:

Word embeddings and LLMs for Genomic Sequences

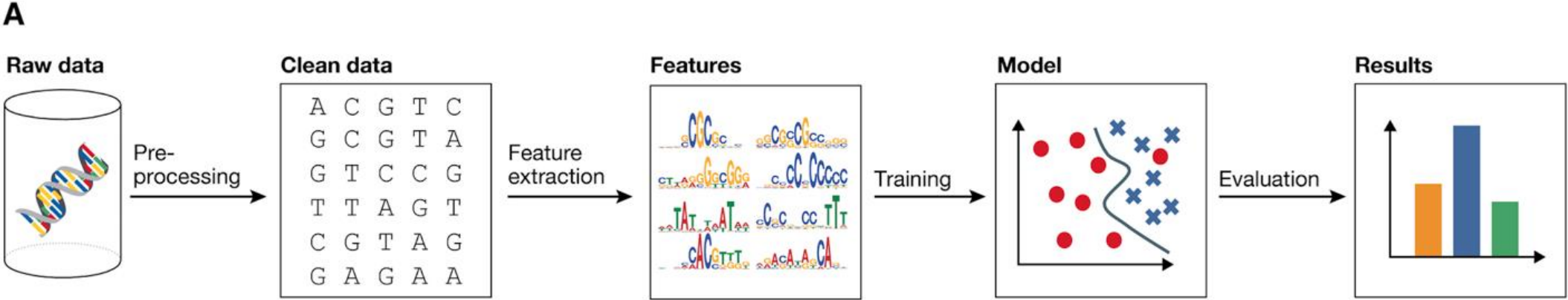
Fabiana Goes

The Rosalind Franklin Institute
Artificial intelligence Group

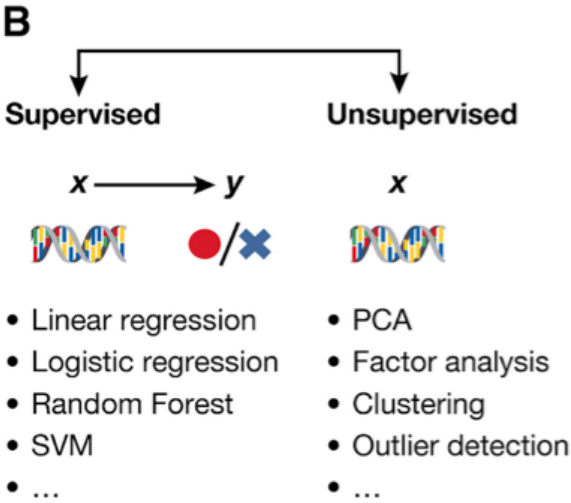
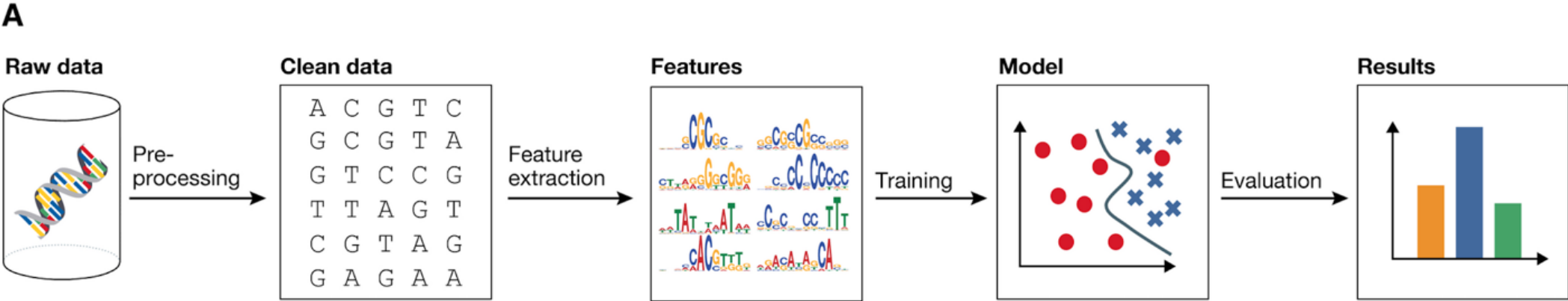
Overview

- Natural Language Processing (NLP)
- NLP and Genomics
- Representation Learning
 - Word2Vec
 - Large Language Models (LLMs)
- Learning goals:
 - How to extract features from DNA sequences using Word2Vec and LLMs

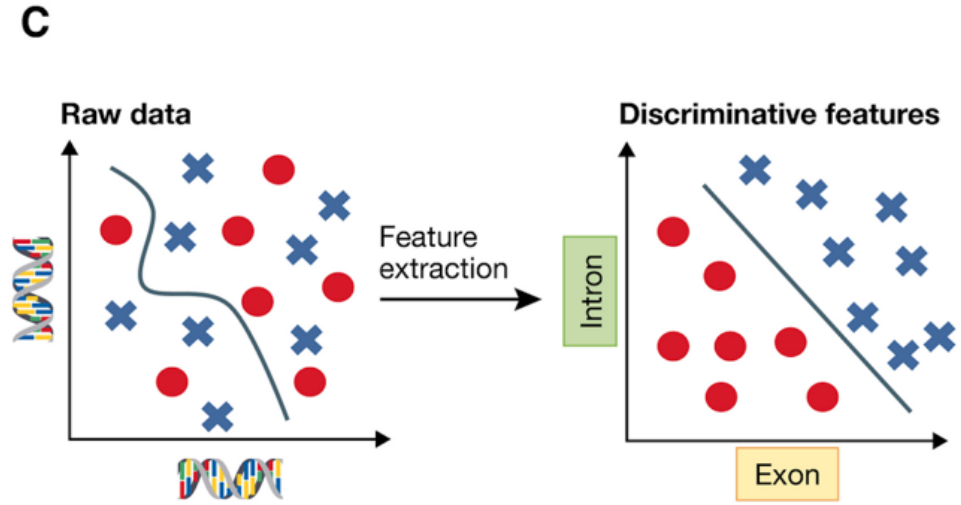
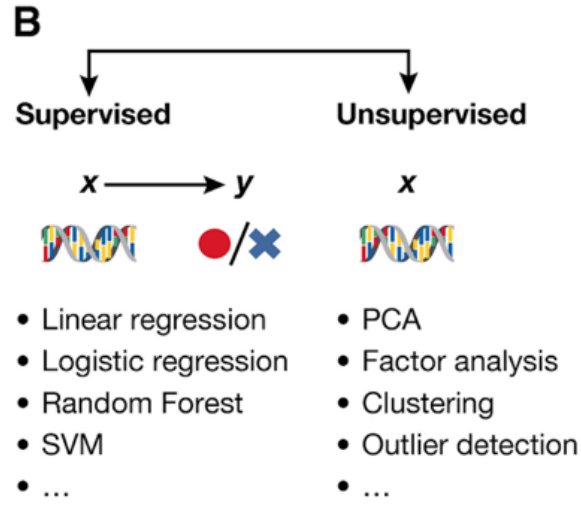
Machine Learning for Computational Biology



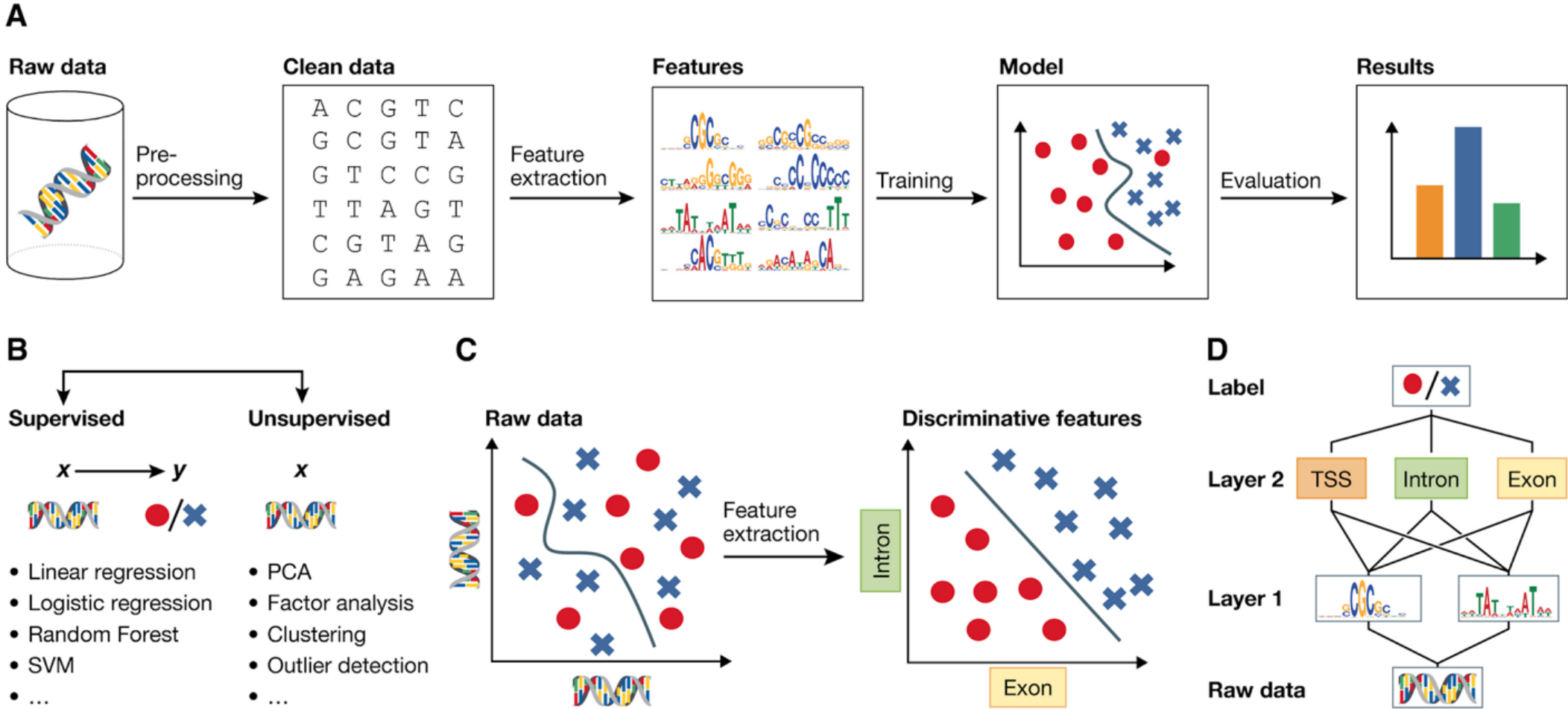
Machine Learning for Computational Biology



Machine Learning for Computational Biology

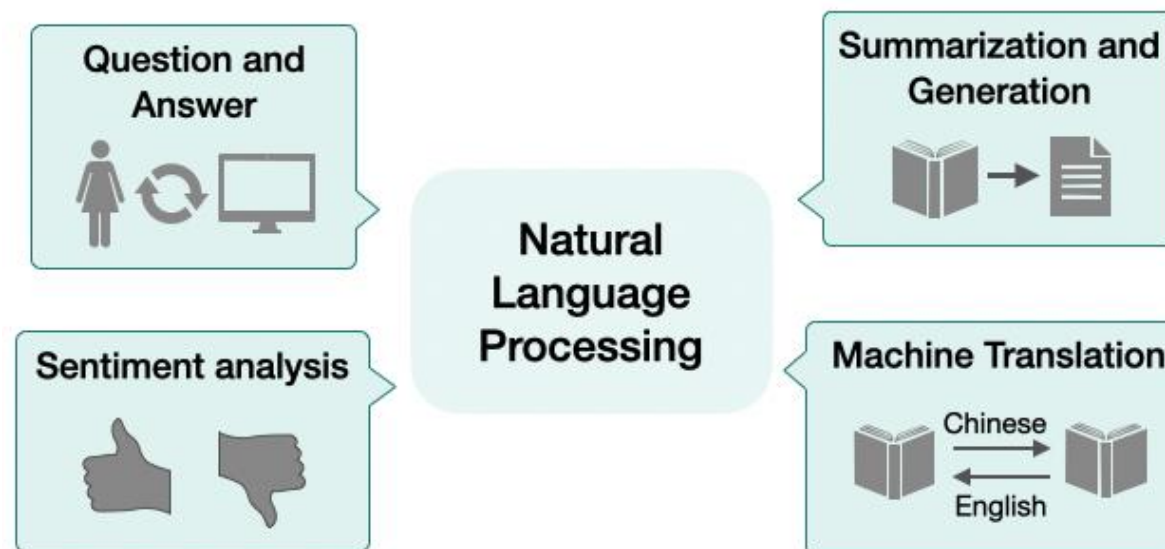


Machine Learning for Computational Biology



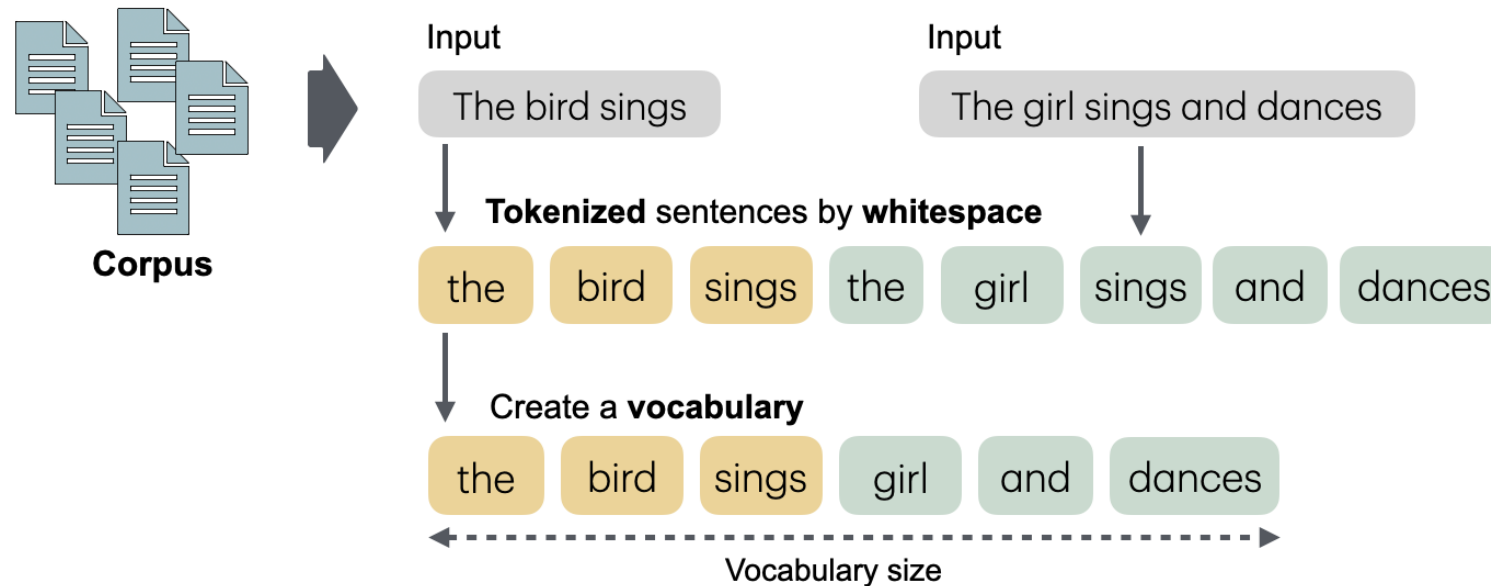
Introduction to NLP

- Natural language processing (NLP)
 - Allow computers to understand the content of natural language
 - To accurately extract information
 - To provide valuable insights
 - Simple linguistic methods and gradually expanded into fields like artificial intelligence and data science



Introduction to NLP - Text representation

- Unstructured data
- Explore the representation of raw data: words or characters
- Converting raw text data into suitable format for computer models
- Tokenization



Introduction to NLP - Text representation

- **Bag-of-words (BoW)**

- Assumption: Texts that use similar words are likely to be similar

Text	good	great	happy	in	bad	terrible	sad	Class
I feel great in this great place	0	2	0	1	0	0	0	1
What a happy day in the city	0	0	1	1	0	0	0	1
Happy news in a good day	1	0	1	1	0	0	0	1
That was a sad and bad experience	0	0	0	0	1	0	1	0
I'm feeling sad in the morning	0	0	0	1	0	0	1	0
I had a terrible day in the school	0	0	0	1	0	1	0	0

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Introduction to NLP

- **Representation learning**

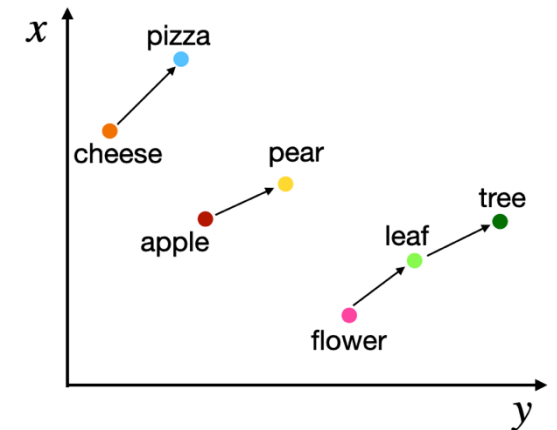
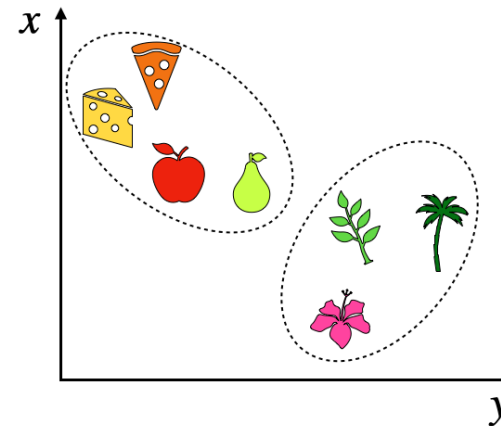
- Models to extract representation from raw data
- Convert words into vectors, preserving their semantic similarity
- The performance of ML depends on the quality of the representation
- Neural networks specialized for **understanding** the **relationships** among **words**

- Embedding models:

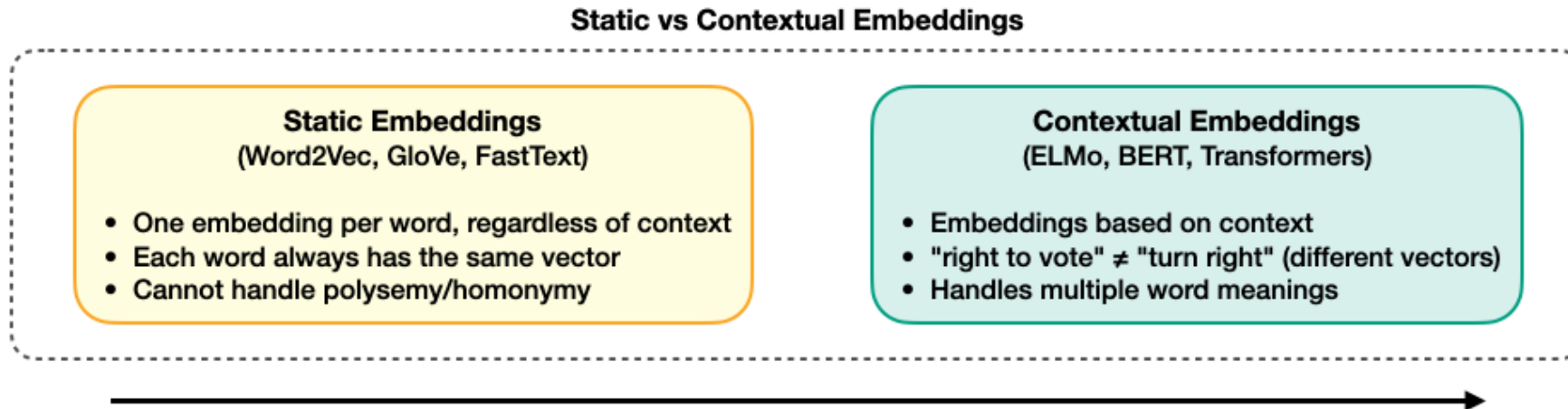
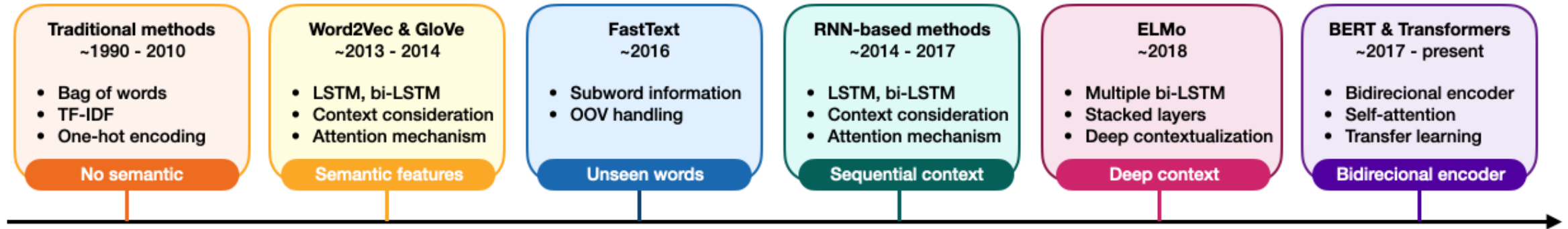
- Transform sequences into numerical vectors capturing patterns
- Assumption: similar words should have similar vector
- High vector similarity between proteins -> similar functions and structures

- Model types:

- **Word2Vec**, Doc2Vec
- CNNs, RNNs
- **Transformer-based architectures (LLMs)**



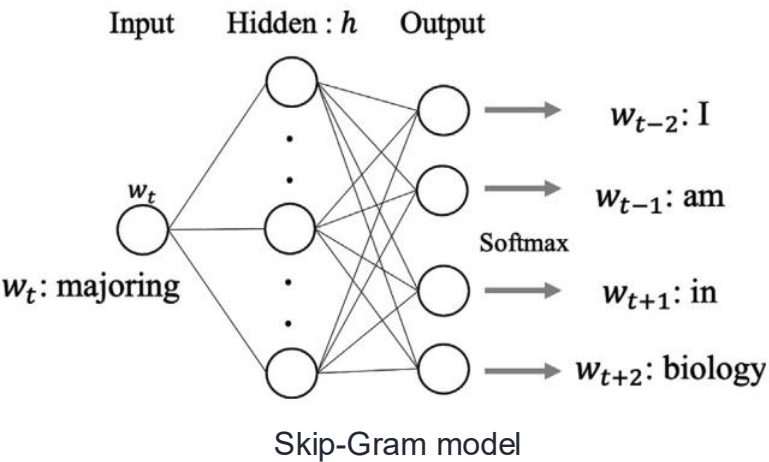
NLP - Timeline of Representation Learning



NLP – Word embeddings

- **Word2vec**: obtain distributed representations using a neural network
 - **Skip-Gram model**: predicts the words around the input word
 - Continuous bag-of-words model: predicts the target word from the surrounding words

Window size	Text	Skip-grams
2	The scientist analyzed data using advanced machine learning	(scientist, the) (scientist, analyzed) (scientist, data)
	The scientist analyzed data using advanced machine learning	(data, scientist) (data, analyzed) (data, using) (data, advanced)
	The scientist analyzed data using advanced machine learning	(machine, using) (machine, advanced) (machine, learning)
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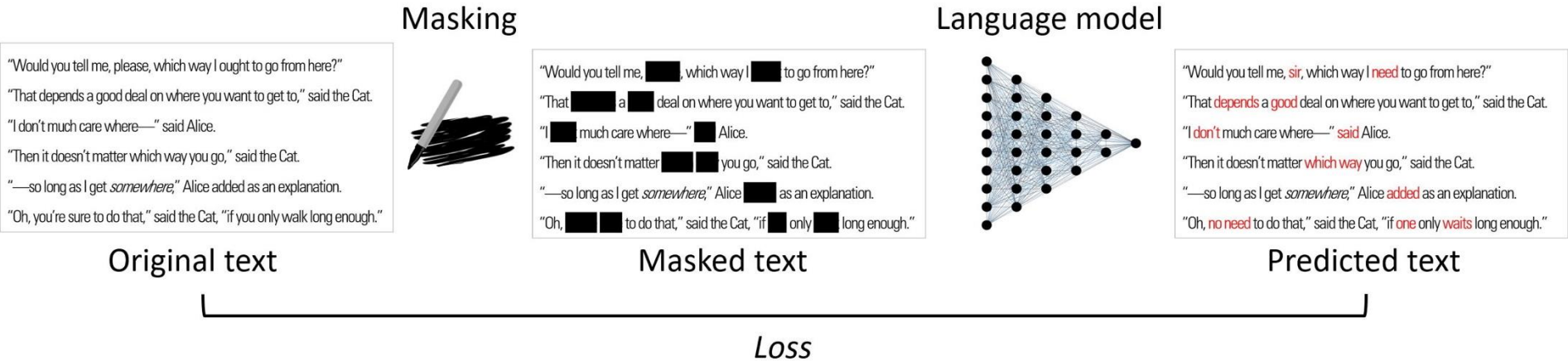


NLP – Deep language models

A Pretraining

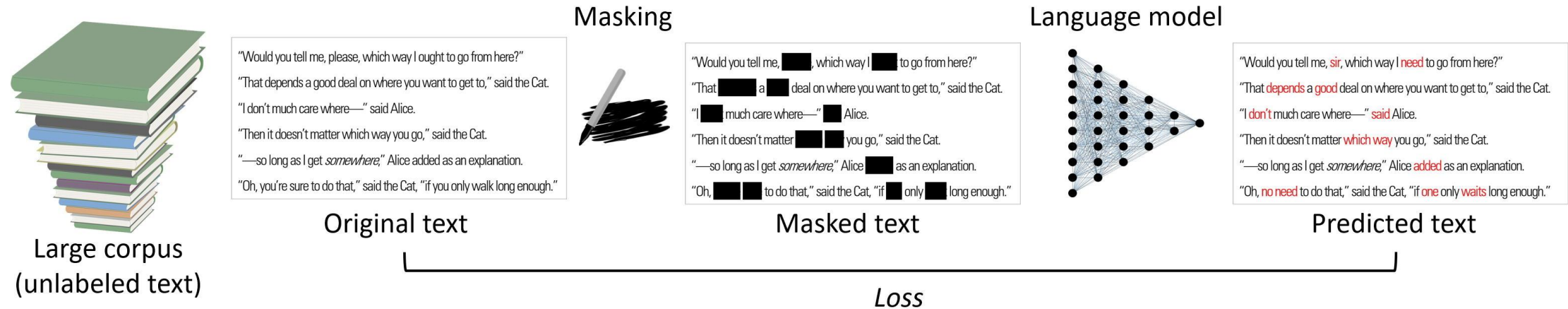


Large corpus
(unlabeled text)

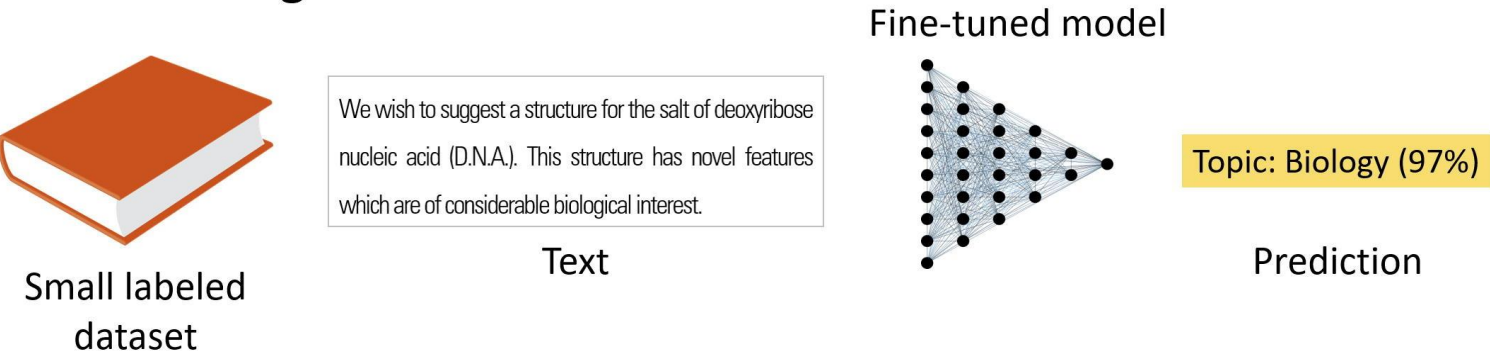


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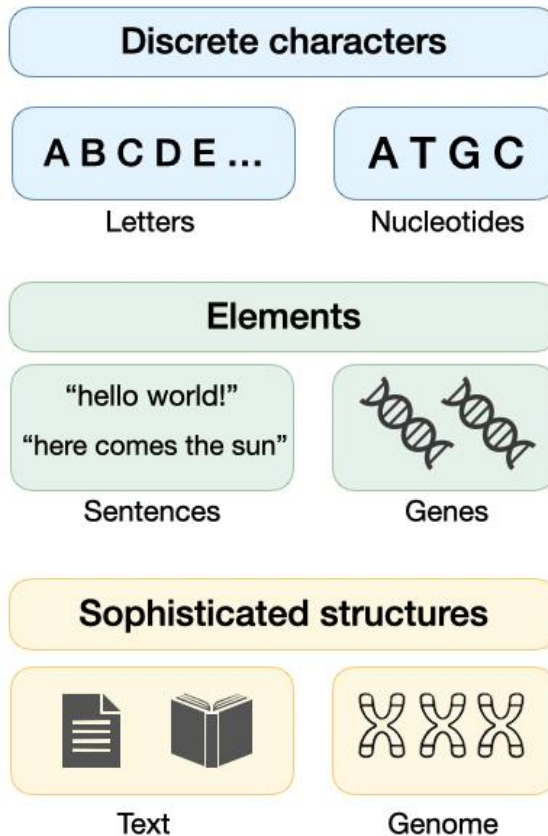
A Pretraining



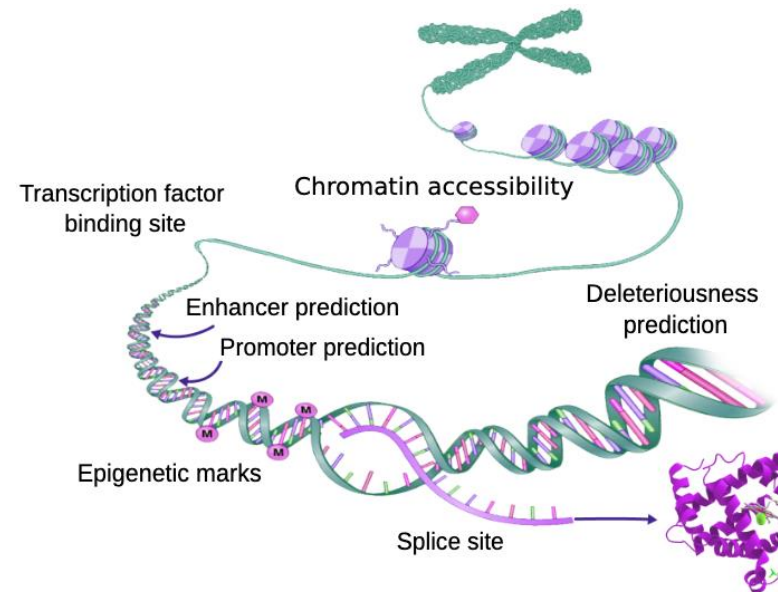
B Fine-tuning



NLP and Genomics

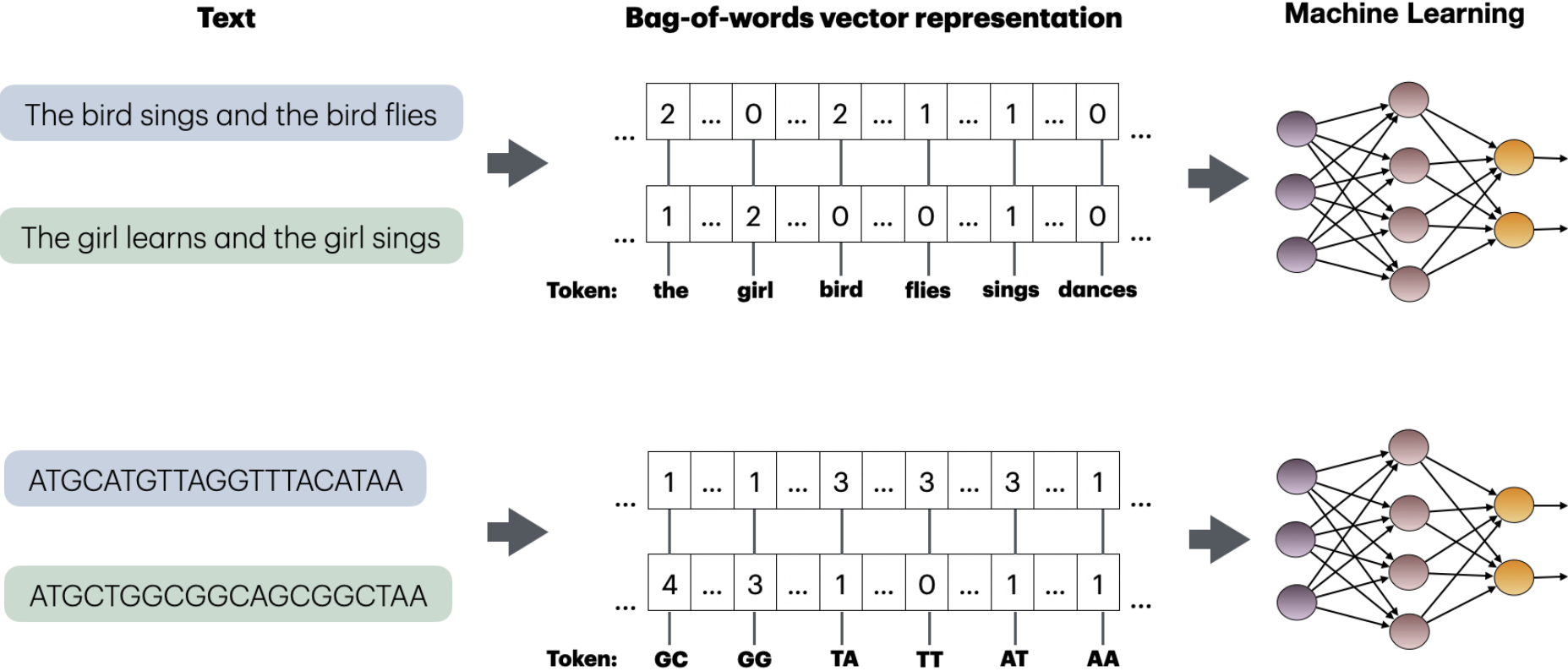


- NLP architectures can be adapted to biological problems
- But...
 - Natural languages: composed by words, each composed of characters from a given alphabet
 - DNA biological sequences: composed of long stretches of nucleotide characters
 - The definition of what is a "word" is **not intuitive**



NLP and Genomics - Tokenization

- Tokenization
 - Sequences are composed of "words" with syntactic and semantic patterns



NLP and Genomics - Tokenization

- Comprehensive understanding of the function and structure encoded in the biological sequence
 - Whether a part of an RNA sequence forms a secondary structure depends on its neighboring sequences
 - The tokenization choice can significantly impact model performance: <https://doi.org/10.1093/bioinformatics/btae196>

a) The baseline “words” tokenizer assumes a dictionary consisting of the nucleotides

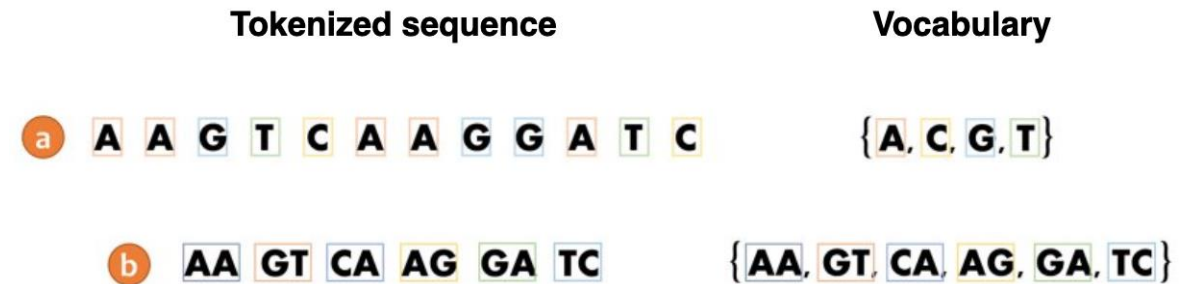
- Small vocabulary
- More long-distance dependencies



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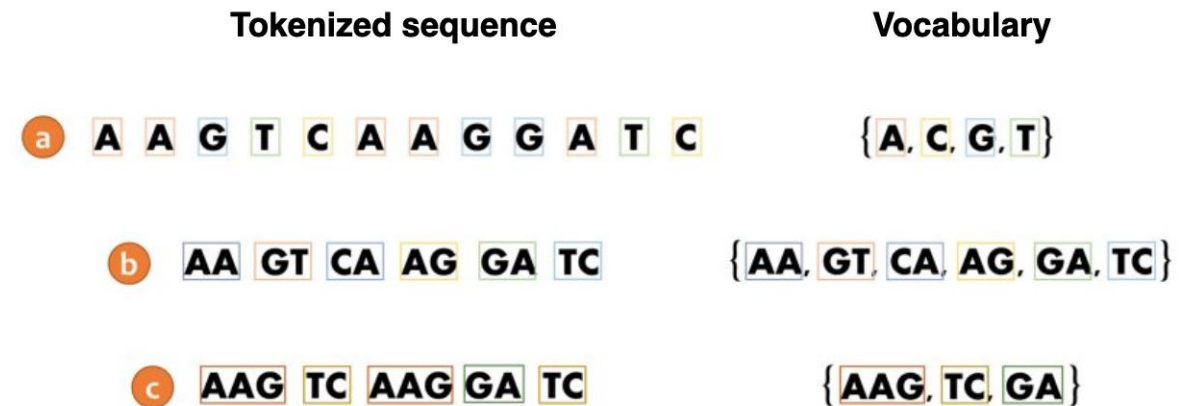
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- Small vocabulary
 - More long-distance dependencies
- b) The “pairs” tokenizer assumes a dictionary consisting of all possible nucleotide pairs



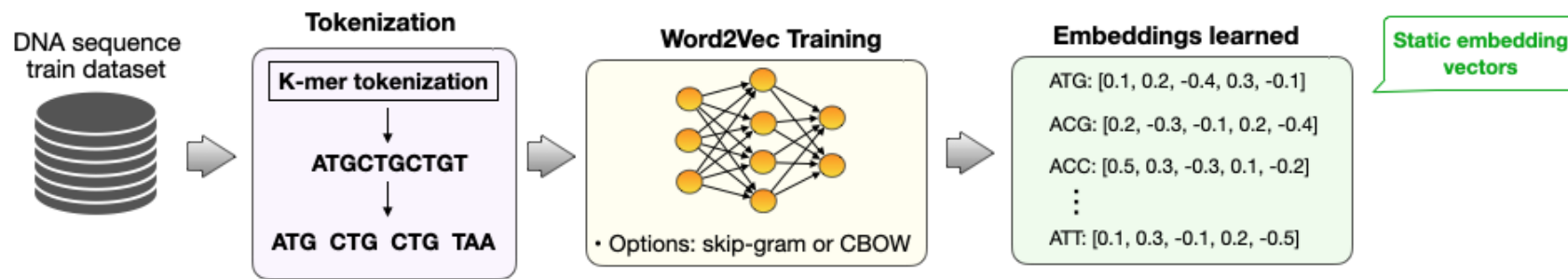
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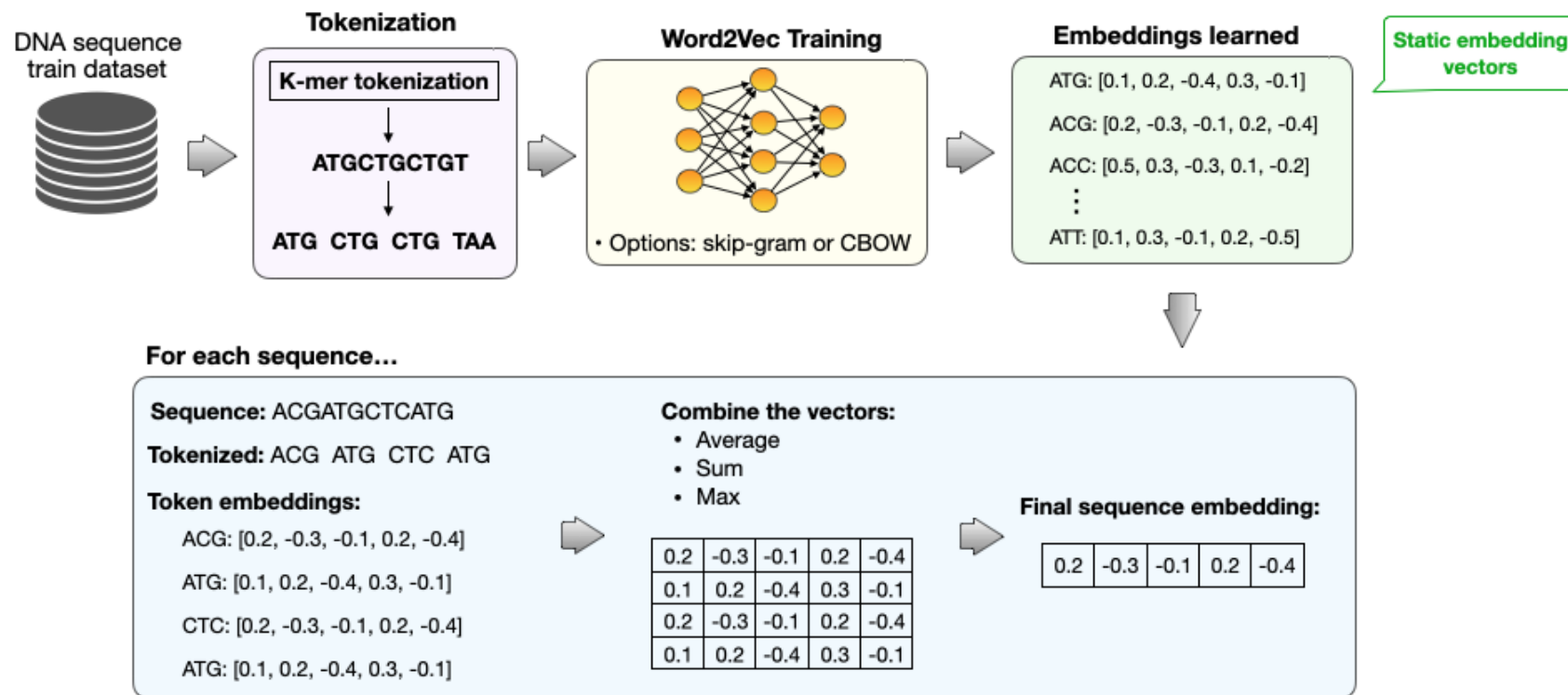
- a) The baseline “words” tokenizer assumes a dictionary consisting of the nucleotides
 - Small vocabulary
 - More long-distance dependencies
- b) The “pairs” tokenizer assumes a dictionary consisting of all possible nucleotide pairs.
- c) Sophisticated tokenizer that consider frequent pairs



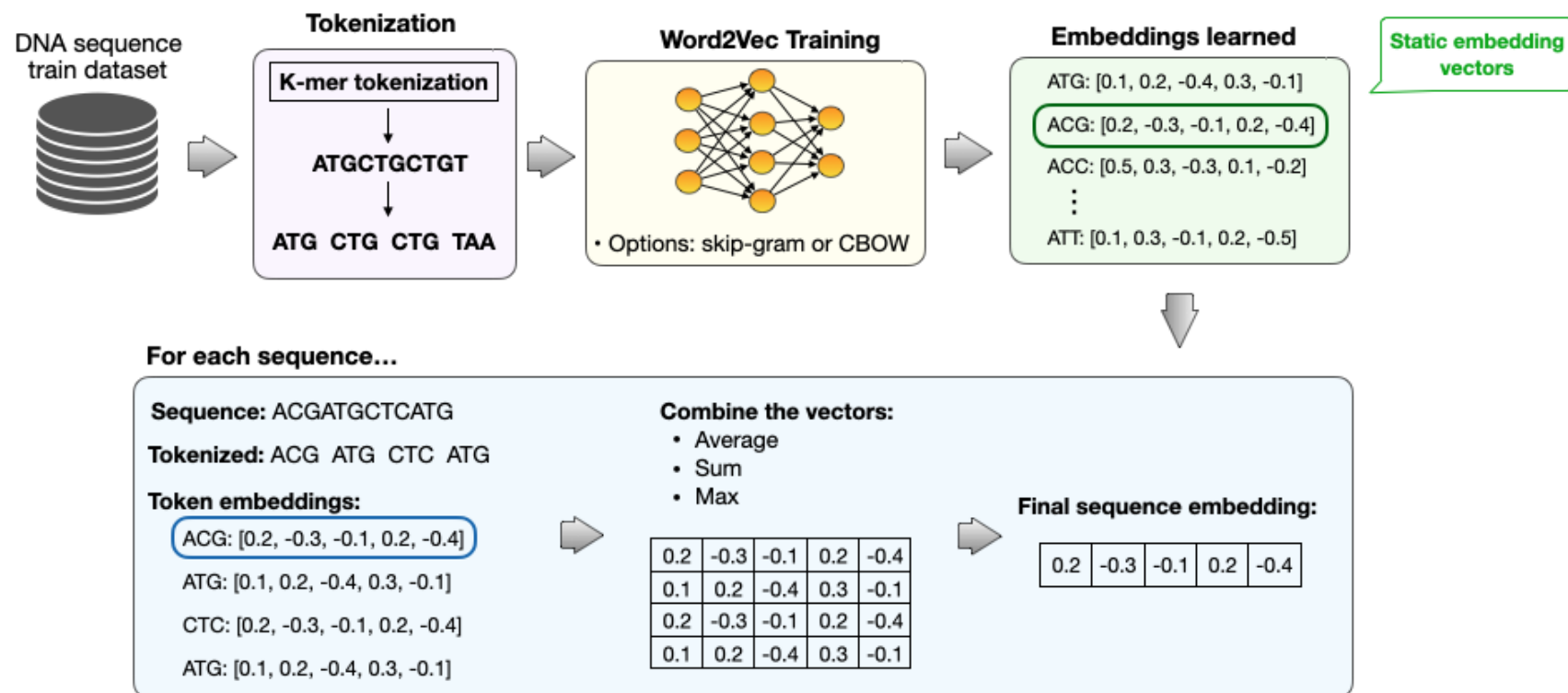
Word Embedding in Genomics



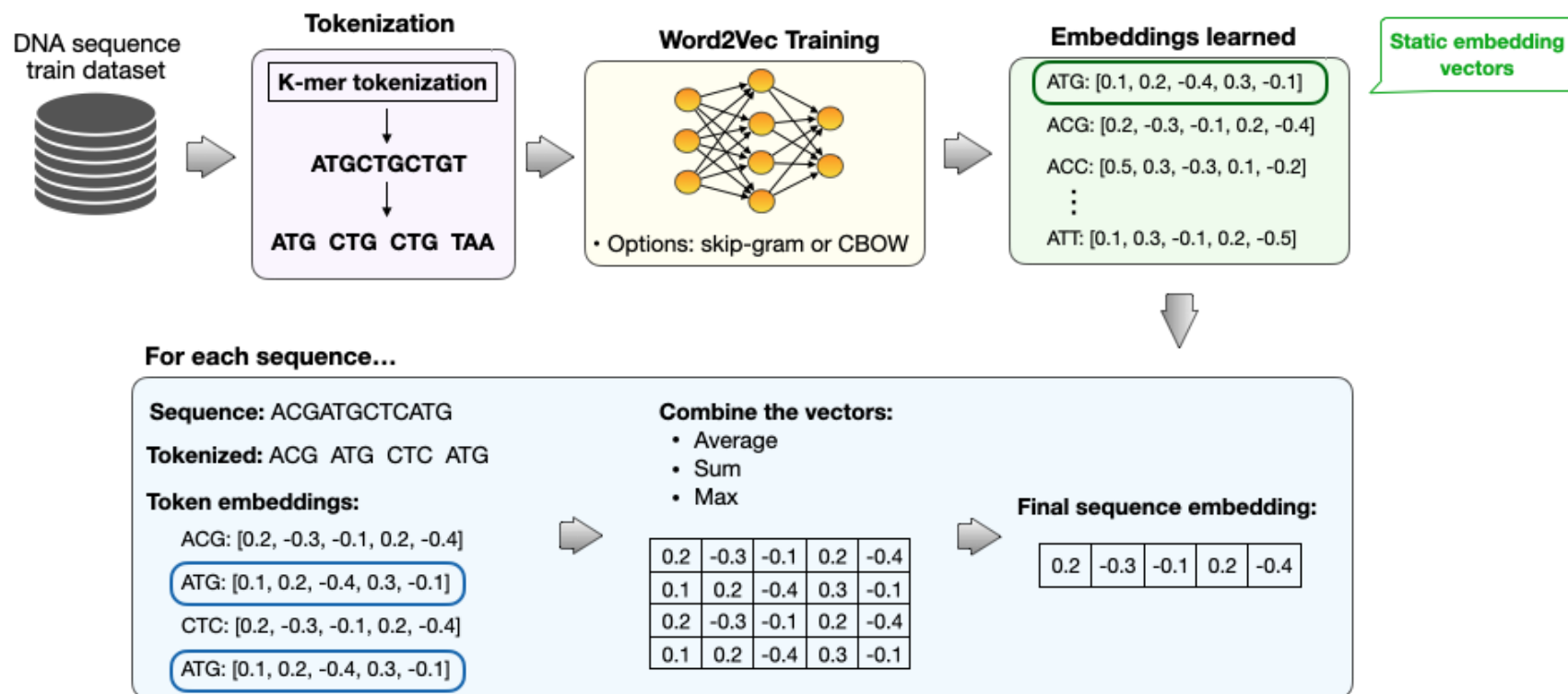
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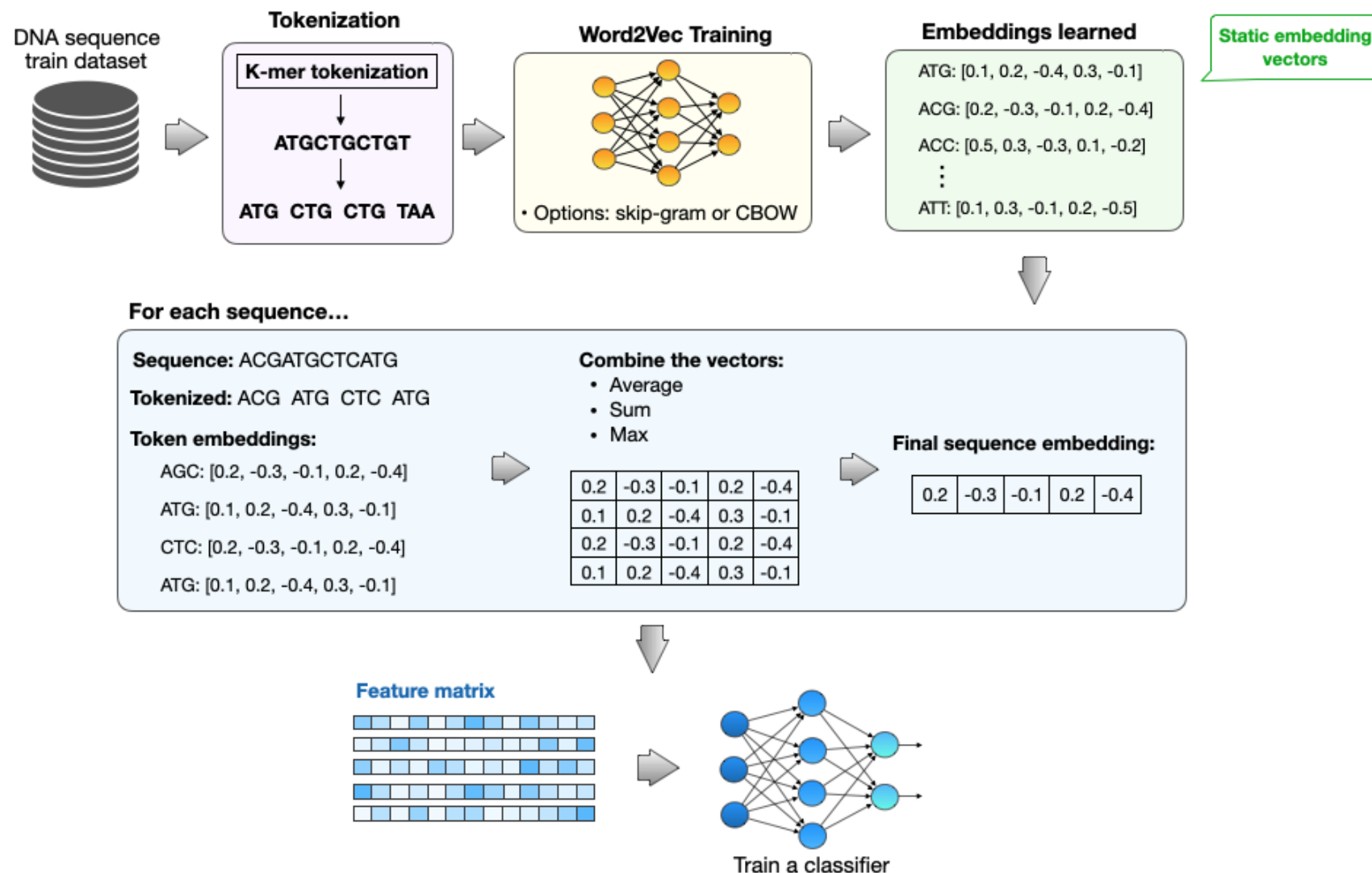
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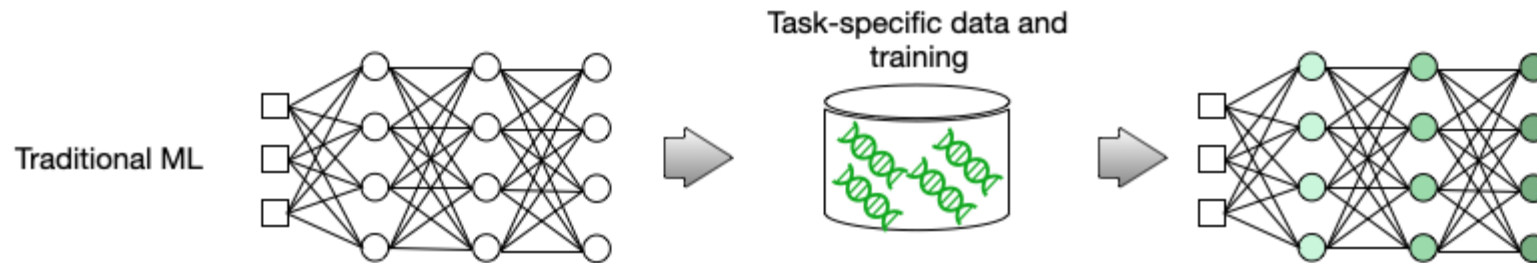


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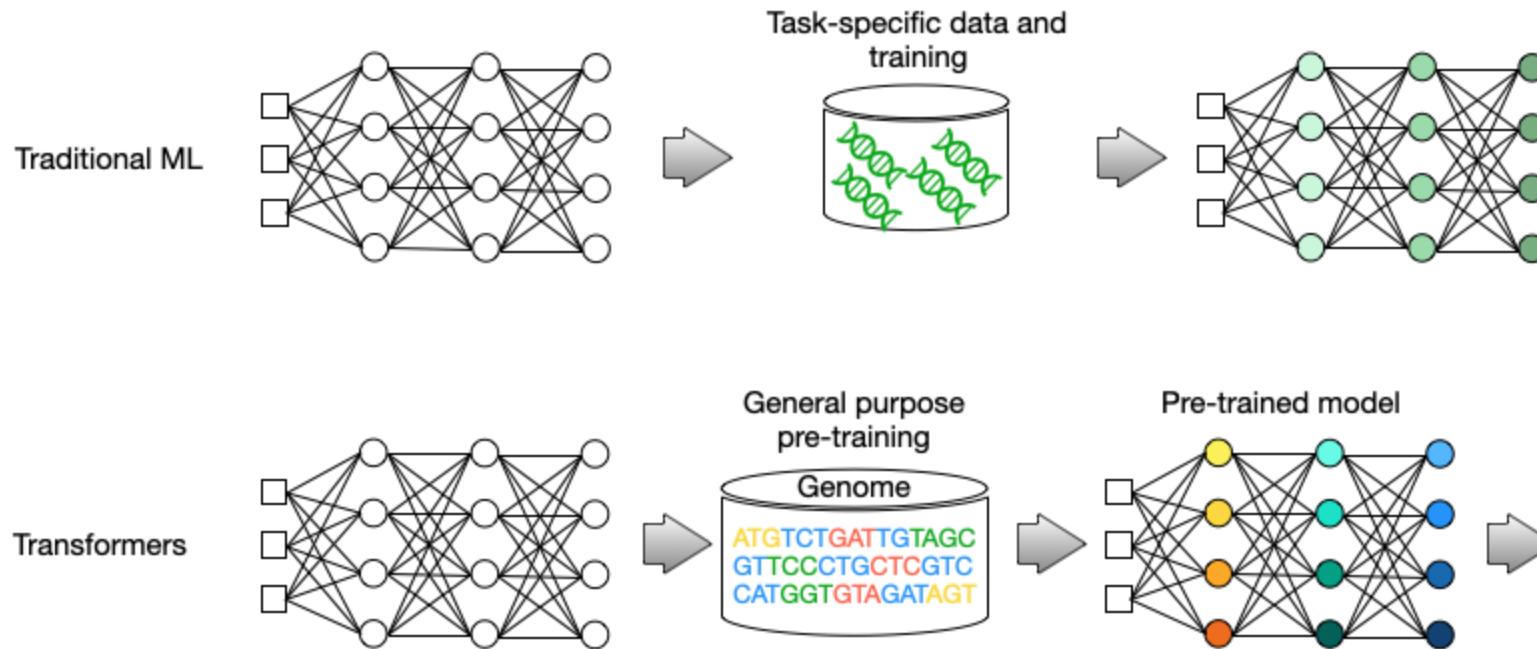
Large Language Models in Genomics

- Learning from vast amounts of unlabeled data
- Capture deep biological patterns, long-range dependencies, and structural information
- Strong capacities to map inputs into latent embedding space
- Can be adapted to a wide range of tasks



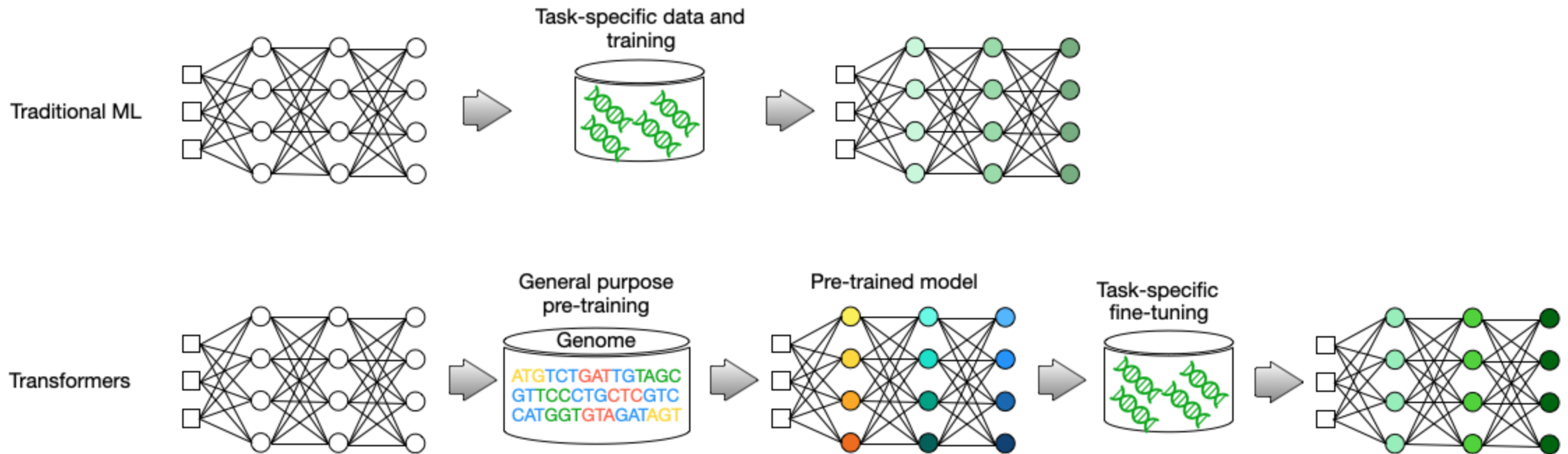
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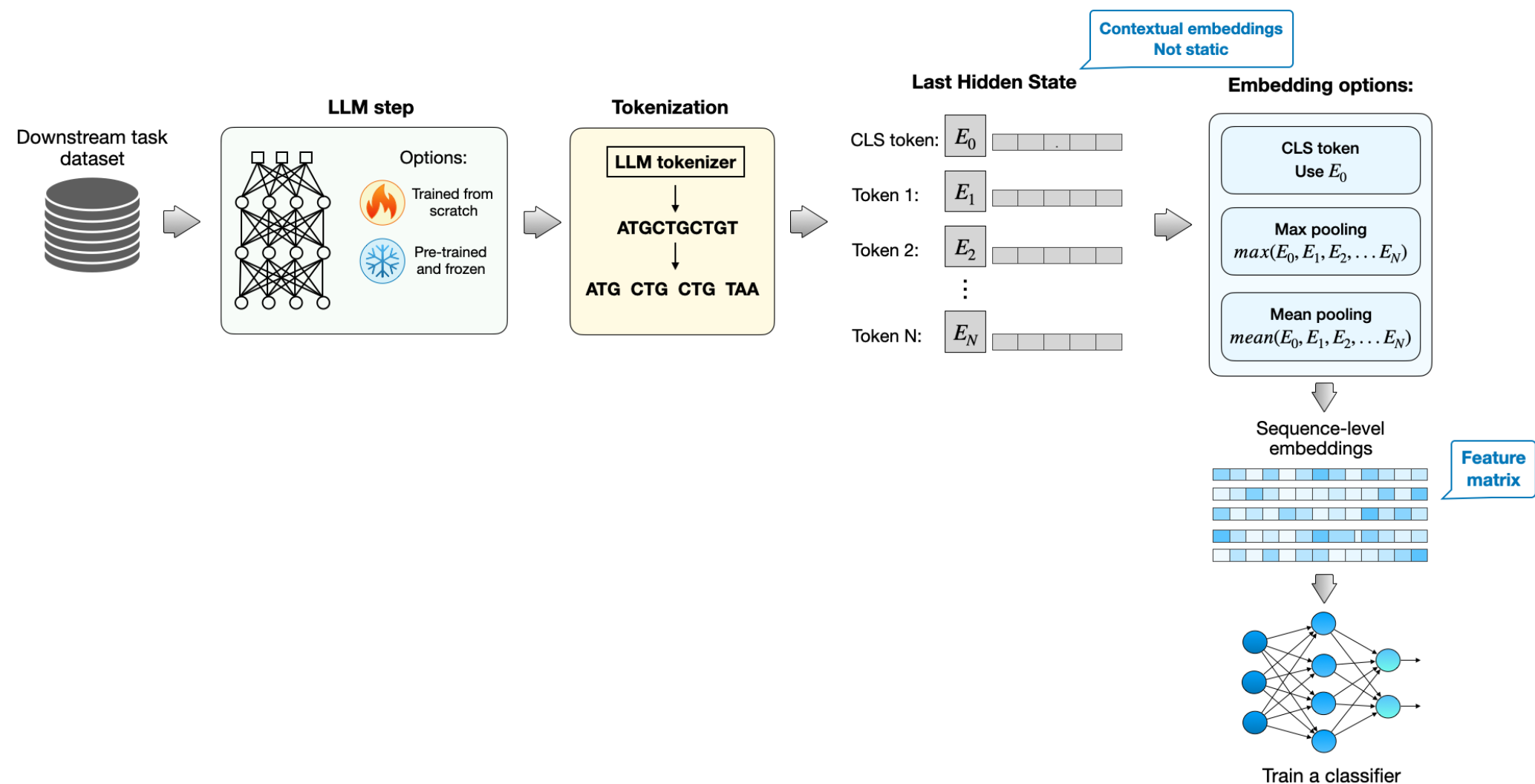
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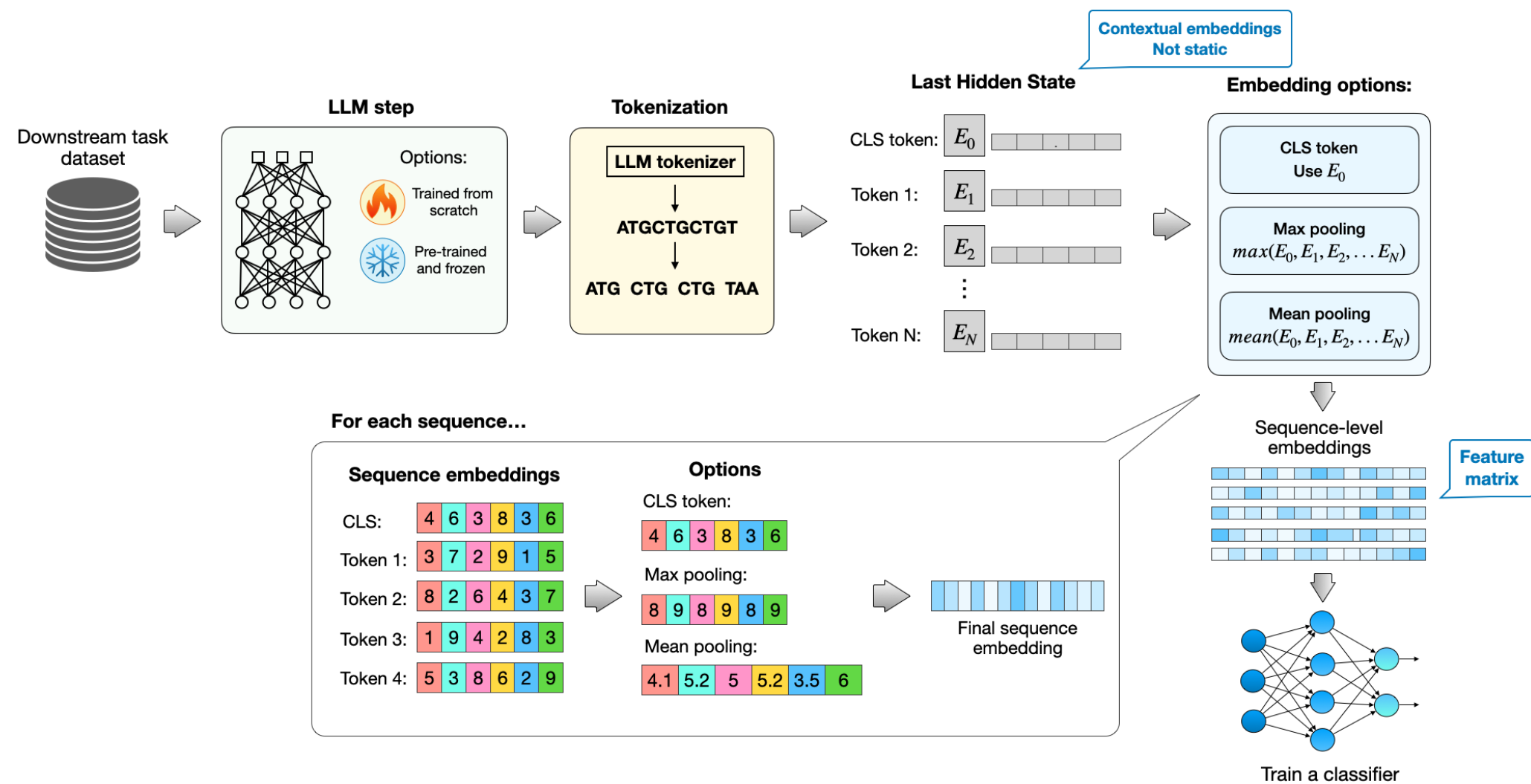
	Sequence Type	Tokenization
DNABERT	DNA	Overlapping k-mer
DNABERT2	DNA	BPE
Nucleotide transformer	DNA	Non-overlapping k-mer
HyenaDNA	DNA	Nucleotide level
EVO	DNA	Nucleotide level
RiNALMo	RNA	Nucleotide level
RNA-FM	RNA	Nucleotide level

- Advantages:
 - Zero-shot and few-shot capabilities
 - Transfer learning across tasks and species
 - Improved performance over traditional models
- Challenges:
 - Computational cost
 - Interpretability
 - Data biases

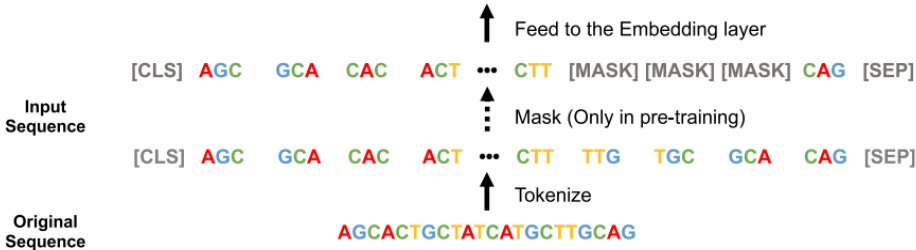
LLM – Pipeline for feature extraction



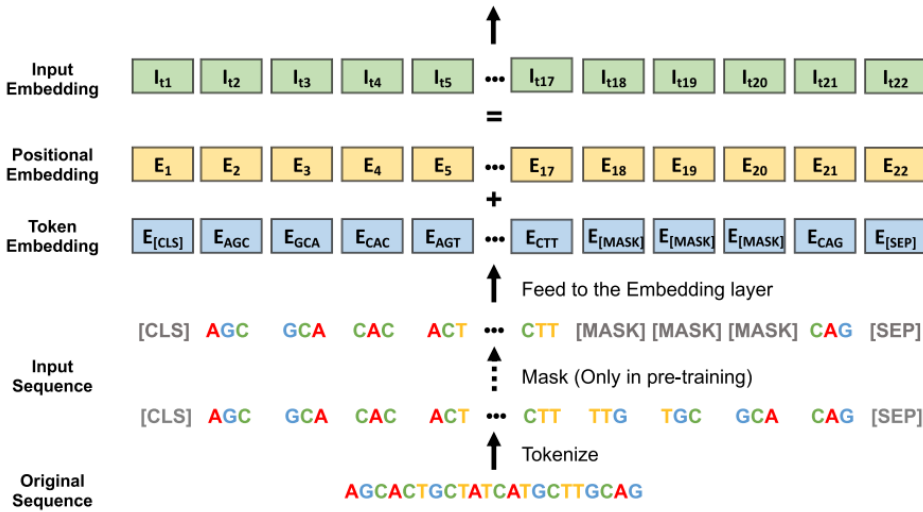
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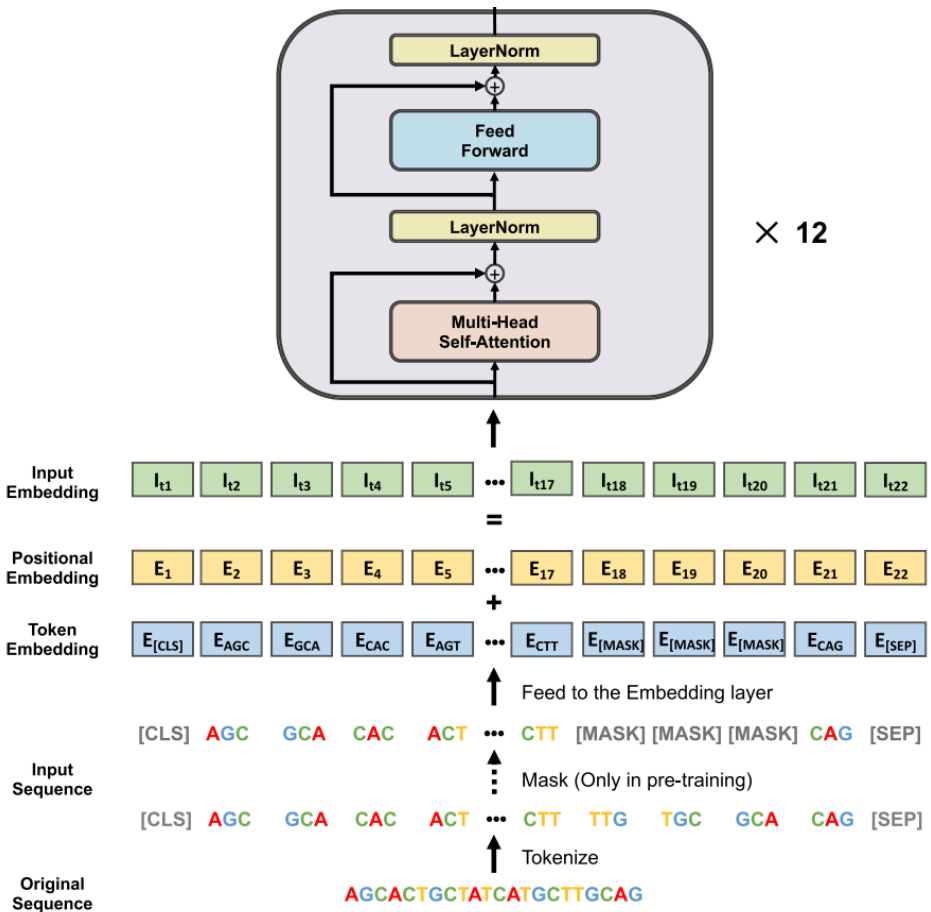
DNABERT – BERT model to DNA



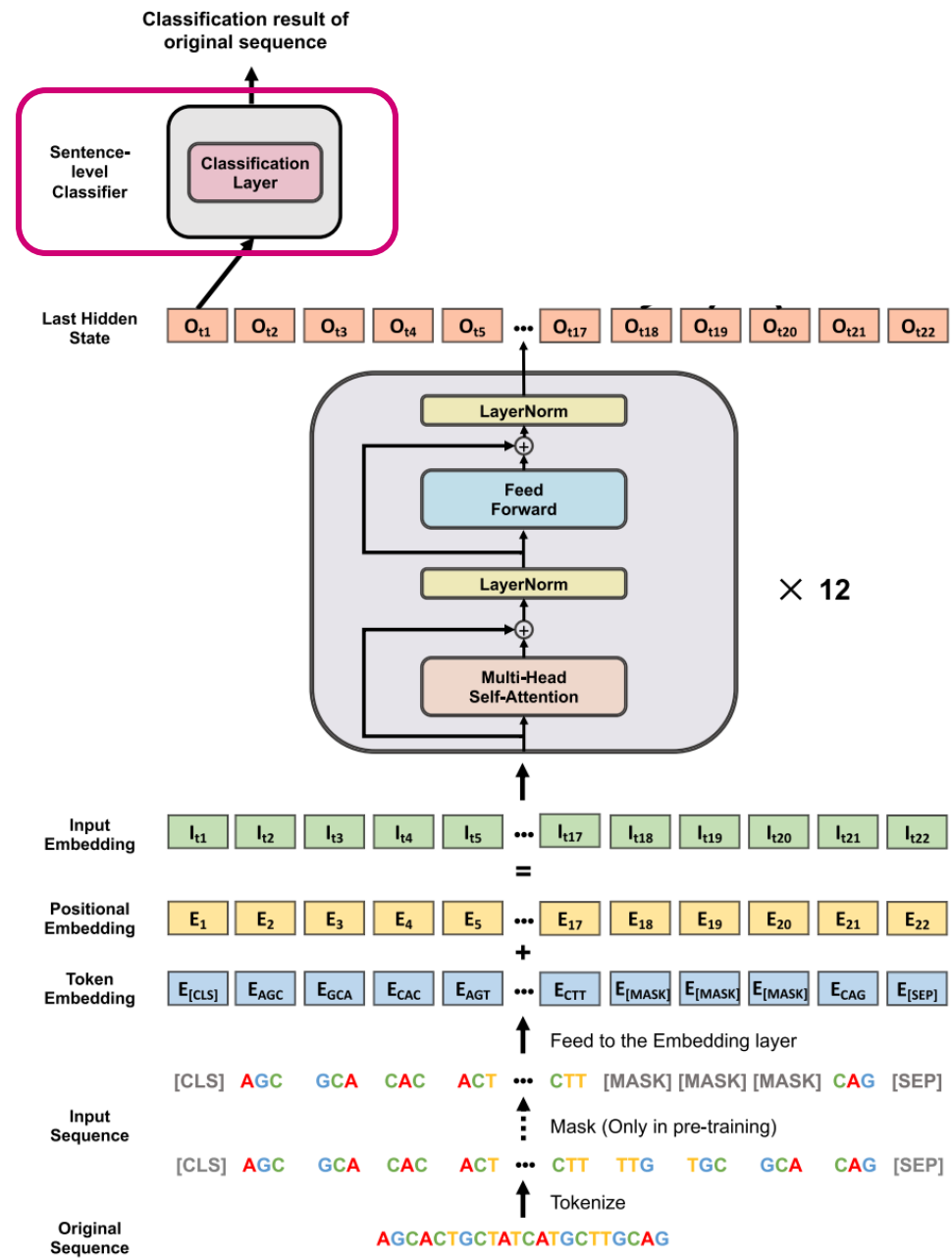
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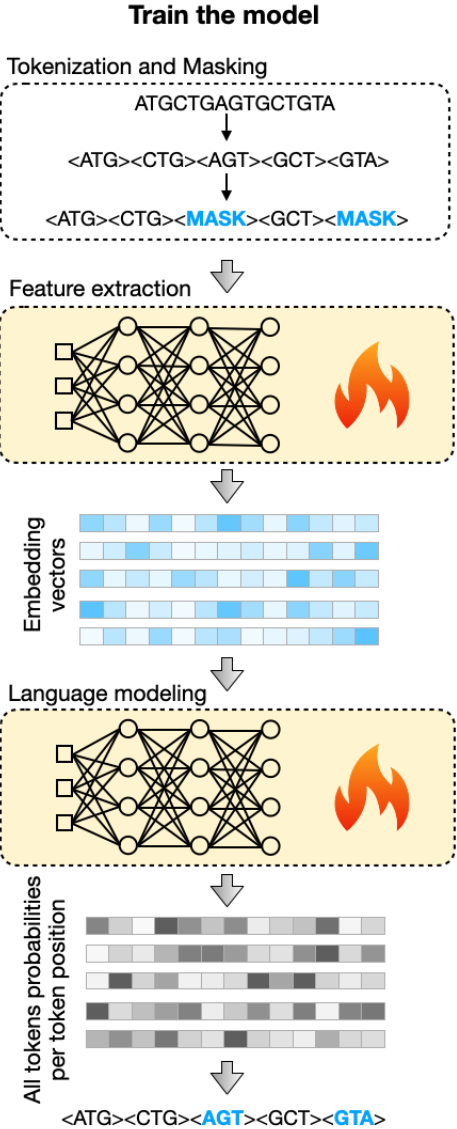
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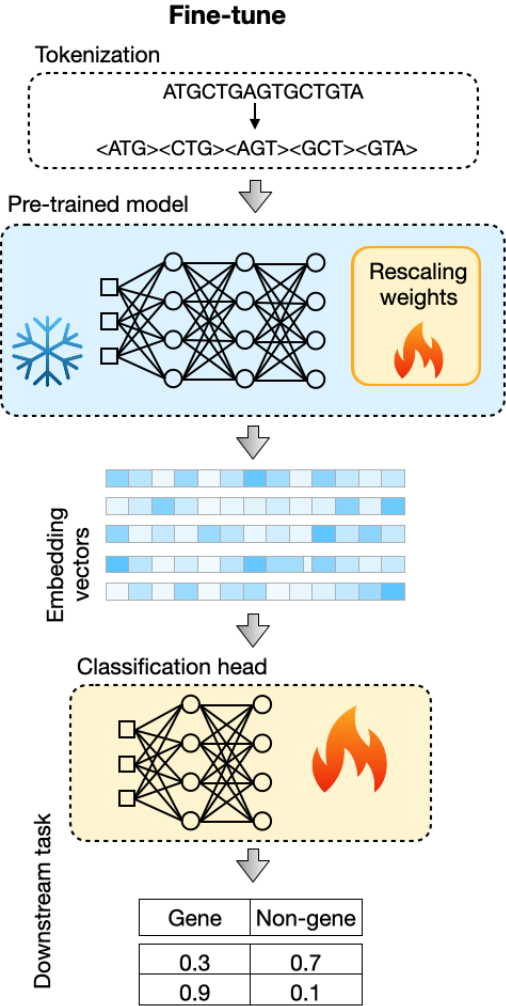
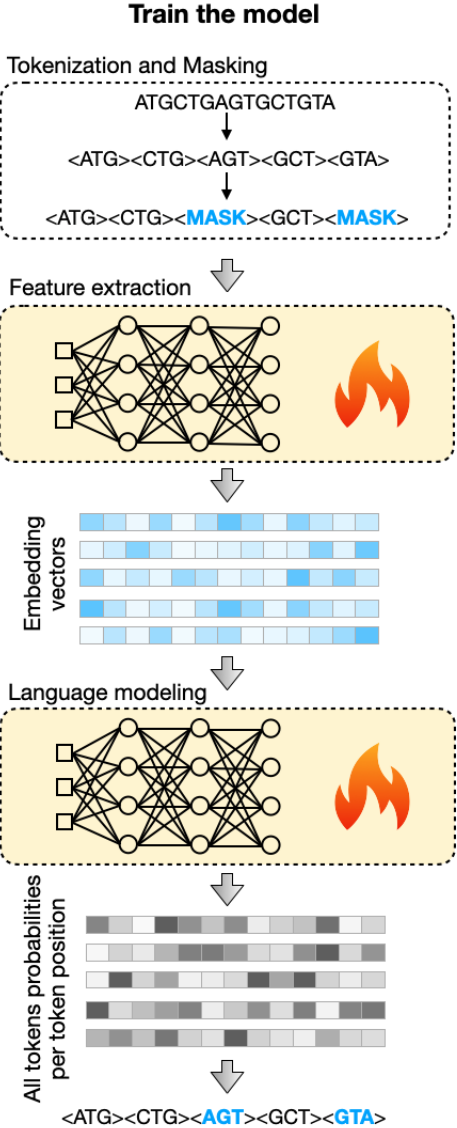


LLMs - Three Approaches



-  Trained from scratch
-  Pre-trained and frozen

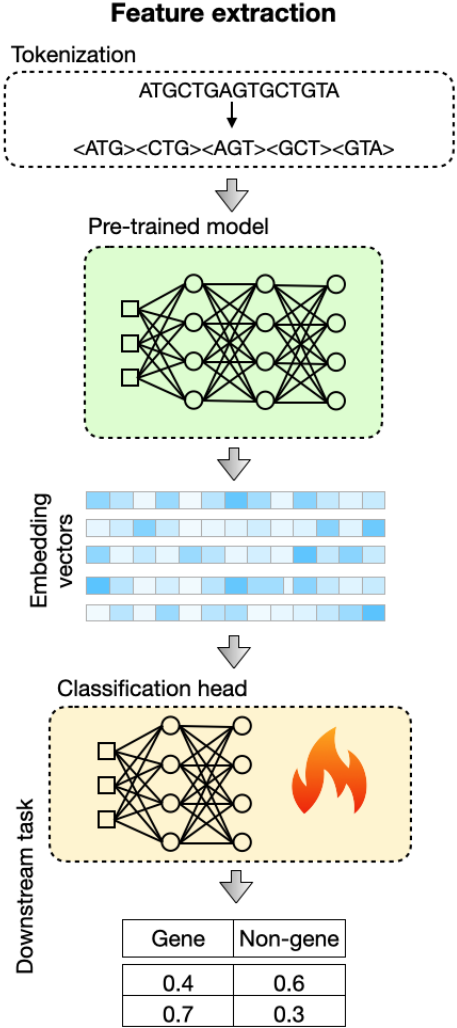
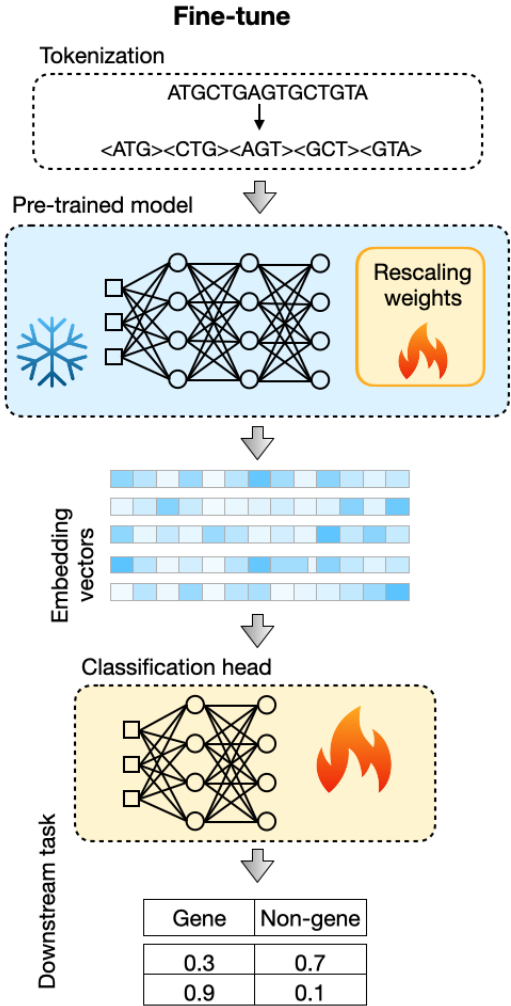
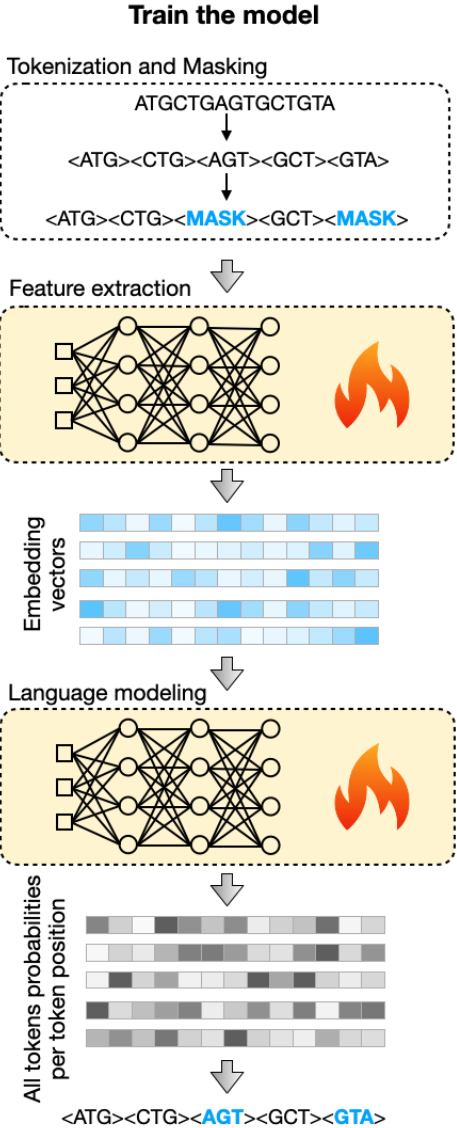
LLMs - Three Approaches



 Trained from scratch

 Pre-trained and frozen

LLMs - Three Approaches



Hands-on: Embedding Genomic Data

From Word Embeddings to Large Language Models

Hands-on

- Practical exercises applying
 - Word2vec
 - LLMs: DNABERT2 and Nucleotide Transformer
- Evaluation and comparison
 - Traditional features and embeddings
- Datasets
 - DNABERT2 benchmark: https://github.com/MAGICS-LAB/DNABERT_2
- Practical code: https://github.com/fabianagoes/ismb_tutorial8

Share your Feedback!

Scan the QR code to let us know your thoughts on this tutorial. Thank you!

